

SAMIKSHA TRIPATHY

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SUMMARY

Computer Science and Engineering undergraduate specializing in Artificial Intelligence and Machine Learning, with hands-on experience in Python, deep learning, computer vision, and audio classification. Builds and benchmarks models end to end, from data pipeline through model evaluation to deployed FastAPI services, and reads research papers to turn ideas into reproducible experiments. Seeking a Computer Vision / AI internship.

TECHNICAL SKILLS

Languages Python, Java, SQL, JavaScript (Basic), HTML, CSS

Machine Learning and Deep Learning PyTorch, TensorFlow, Scikit-learn, Convolutional Neural Networks (CNN), Transfer Learning, Transformers, Embeddings, Supervised Learning, Classification, Regression, Random Forest, Model Evaluation and Benchmarking

Computer Vision Image Classification, Fine-Grained Visual Classification, Data Augmentation, Torchvision, OpenCV, Pretrained Backbones (ResNet, EfficientNet), Object Detection (Foundational)

Audio and Signal Processing Audio Classification, Mel-Spectrogram Analysis, Librosa, Torchaudio, SpecAugment

AI Engineering Prompt Engineering, Large Language Model (LLM) APIs, AI-Assisted Development, Recommendation Systems, Natural Language Processing (NLP)

Data and Analysis Pandas, NumPy, Matplotlib, Jupyter Notebook, Feature Engineering, Experiment Design

Backend and Tools FastAPI, REST APIs, PostgreSQL, MySQL, SQLAlchemy, Pydantic, Docker, Git, GitHub, pytest, Linux, VS Code, Hugging Face Spaces

PROJECTS

Bird Species Identification from Images and Audio | *PyTorch, Torchaudio, Librosa, FastAPI* 2026

- Built a multimodal deep learning system that identifies bird species from both photographs and recorded bird calls, using a fine-tuned convolutional neural network for images and a mel-spectrogram CNN for audio.
- Fine-tuned a pretrained image backbone on a fine-grained bird dataset, applying data augmentation to improve robustness to varied angles, lighting, backgrounds, and partially visible subjects.
- Converted Xeno-Canto field recordings into mel-spectrograms and trained an audio classifier, benchmarking accuracy against the open-source BirdNET baseline.
- Deployed the system as a public web demo with a FastAPI inference endpoint returning top-3 species predictions with confidence scores.

AgriOptima AI — Crop Recommendation Engine (Backend Lead) | *Python, FastAPI, PostgreSQL, Docker* 2026

- Designed and built the entire backend for a crop recommendation platform advising farmers on viable crops from soil, irrigation, season, budget, and location inputs.
- Built the soil and agronomic eligibility engine from scratch, encoding deterministic rules for soil pH, water availability, season, budget, and rotation that reject unsuitable crops with explainable reason codes before scoring.
- Implemented a transparent 0-100 scoring model across soil fit, climate fit, water fit, demand, price, and saturation risk, returning ranked recommendations with score breakdowns and confidence levels.
- Integrated live external data (Open-Meteo weather, data.gov.in AGMARKNET mandi prices) through adapters with timeout handling and labelled cached fallbacks, so no output is presented as more certain than its source.
- Connected all components through a FastAPI service layer with Pydantic validation, SQLAlchemy and Alembic over a Dockerized PostgreSQL database, and an automated pytest suite.

Cache Eviction Policy Benchmark: LRU vs LFU vs Machine Learning | *Python, Scikit-learn* 2026

- Designed a controlled research benchmark comparing a learned eviction policy against LRU and LFU baselines on a synthetic Zipfian trace with an injected scan to test scan resistance.
- Trained a 200-tree Random Forest regressor on oracle reuse-distance features as an offline approximation of Belady's algorithm, with a strict train/test split to prevent data leakage.
- Measured hit rate and per-eviction latency across a cache-size sweep: the model matched LFU on hit rate (0.559 vs 0.561) while running roughly 14,000x slower per decision, documenting when learned eviction is worth its cost.

EDUCATION

SRM Institute of Science and Technology *Kattankulathur, Tamil Nadu* Expected Aug 2028

B.Tech, Computer Science and Engineering — Artificial Intelligence and Machine Learning Specialization | GPA: 9.34 / 10

Relevant Coursework: Machine Learning, Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks

CERTIFICATIONS, LEADERSHIP AND INTERESTS

- Supervised Machine Learning: Regression and Classification — DeepLearning.AI and Stanford Online (Coursera), Andrew Ng.
- Community Volunteer, multiple local NGOs (Ongoing) — contributed time to community-focused initiatives and social outreach.
- Interests: Birding and wildlife photography, digital art and painting, graphic design, UI design.